EFU-AI / 200.1

Biophysical Decision Architecture: Residential Building Retrofit MROI Assessment

Specification Working Paper — v1.0

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Date: 2026-02-27
Author: Simor, István
ORCID: https://orcid.org/0009-0002-6599-3480
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Abstract

Current AI decision support systems are built on static category logic and text-based training data, which implicitly reinforce existing economic and infrastructural distortion patterns (600.69 Gresham Spiral). This document specifies the first operational module of the EFU-AI system's 200-series: a hybrid (rule-based + regression) decision architecture validated on a residential building retrofit assessment case study (905.0 PILOT). The base unit is 1 EFU = 20 kg/day metabolic flux. Decision outputs are based on HMI, JIM-30, SSF and MROI metrics, coordinated with the CEWS macro-level early warning system. The system's goal is not categorization but flux-consistency verification and early detection of antiflux patterns. The ATP-1.0 falsification protocol ensures scientific testability: pilot results (DOI: 10.5281/zenodo.18789313) confirmed that decision outcomes remain stable under ±20% weight perturbations. The v1.0 document explicitly records both empirically calibrated and pilot-data-pending parameters — notation convention: [empirical] vs. [estimated — 905.0 validation pending].

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1. Introduction and Problem Statement

1.1 Three Structural Limitations of Current Decision Support

Category bias. Training data encodes patterns of past human decisions, including CAPEX-minimizing, GDP-oriented and short-term optimization biases. The system treats these as implicit norms and reproduces them in every decision cycle.

Consensus lag. Academic consensus and regulatory frameworks lag behind physical reality. A system trained on 2015 literature consensus produces faulty decision inputs in 2026, particularly in rapidly evolving energy contexts where grid carbon intensity and technological maturity change annually.

Antiflux blindness. The patterns of the 600-series — lock-in, regulatory capture, hype cycles, Gresham Spiral — do not appear in standard training data, because that data is precisely a description of the existing system. What the system learns as normal, it cannot flag as distortion.

1.2 The EFU-AI Approach

EFU-AI operates on flux vectors, not text data. The fundamental question in every decision situation is not which category does this system belong to, but: what flux pattern does it exhibit, and where is it heading?

This ontological shift has three operational consequences, which this document formalizes:
(1) rule-based threshold filtering on physical constraints,
(2) regression refinement with antiflux correction,
(3) macro-level CEWS feedback that places project-level decisions in a system-level risk context.

1.3 Scope and Position of This Document

This document specifies the first decision situation of the 905.0 PILOT — residential building retrofit MROI assessment — with a complete architecture. The system is calibrated to Hungary's 2024–25 grid parameters (RACF = 526 kg CO_2/person/year, grid_CO_2 = 0.35 kg/kWh, MAVIR), but the methodology is transferable to other contexts with different RACF values.

Within the EFU ontological structure, this document is the first specification module of the 200-series (Digital Backbone). It connects to the 900.6.1 Operational Manual, the 600.69 Gresham Spiral diagnostic description, and the 700.13 CEWS unit.

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2. The EFU Base Unit and Key Metrics

2.1 The EFU Base Unit

1 EFU = 20 kg/day metabolic flux, comprising three components:

• 3 kg water
• 0.5–2 kg biomass
• 15 kg oxygen

This is the scale-invariant biophysical base unit on which the entire assessment system is built. Full derivation is available in MROI Working Paper v1.3 (DOI: 10.5281/zenodo.18766850).

2.2 HMI — Useful Work Index

The HMI measures the metabolic efficiency of the retrofit intervention on a 0–1 scale:

HMI Value	Category	Decision Consequence
< 0.2	VETO	Project halt
0.2–0.5	RESTRICTED	Conditional continuation
0.5–0.8	STABLE	Standard assessment
> 0.8	SYMBIOTIC	Prioritized assessment

2.3 JIM-30 — Minimum Repairability Index

JIM-30 consists of four equal dimensions, 0–25 points each:

• Modular design
• Parts availability
• Documentation quality
• Local repairability

Aggregate threshold: < 20% = automatic VETO — the regression layer does not run.

2.4 MROI — Metabolic Return on Investment

$$MROI = (\Sigma\ \Delta E_saved \times grid_CO_2 / RACF) / \Sigma\ EFU_input(corrected) \times 100$$

RACF — Reference Annual Carbon Flux
Hungary: 526 kg CO_2/person/year [empirical — MAVIR 2024–25]
Grid carbon intensity: 0.35 kg CO_2/kWh [empirical — MAVIR 2024–25]

2.5 SSF — Social Flux Index

Full description in Section 4. The SSF measures the social labor flux impact of the retrofit intervention across three dimensions: repair labor time, local competence, and supply chain independence.

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3. The EFU-AI Hybrid Decision Architecture

3.1 System Diagram

INPUT VECTOR
[ΔE_saved, grid_CO_2, EFU_input, JIM-$30_{1–4}$, HMI, $SSF_{1–3}$]

LAYER 1: Rule-based filter
• HMI threshold check
• JIM-30 threshold check
• MROI_preliminary calculation
Output: VETO / RESTRICTED / STABLE

↓ Passed

LAYER 2: Regression refinement
• $\beta_1 = –0.12$ [empirical]
• β_2, β_3 [estimated]
• SSF (X_3) integration
• ε_antiflux^total computation
• CEWS feedback (level 0–3)

↓

OUTPUT
• Decision recommendation
• MROI_final
• Severity level (217.1)
• CEWS level (700.13)
• Flux audit summary
• Alternative MROI solution

State Transfer Block — v1.0 Final

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β_1: –0.12 [empirical]
β_2, β_3: [estimated — longitudinal validation 2027–2030]
SSF weights: 0.4 / 0.3 / 0.3 [justified]
Severity: B: 2.1 | A: ~3.0
CEWS level: 1 — Warning, additive term active
Antiflux: 600.xx negative (project-level, Scenario B)
ATP-1.0 F1: Not activated
Next module: 200.2 — Antiflux Filter PILOT #2/#3
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